

Held fixed across the four primary arms

256 deterministic master keys; accepted seed matrices S_j ; round keys; 128-bit state-rotation schedule; AES S-box; byte-routing permutation; input panels



Only experimental variable: orientation exponent $e(r, j)$ in $R^{e(r, j)}(S_j)$

static

$$e(r, j) = 0$$

position-only

$$e(r, j) = j \bmod 4$$

round-only

$$e(r, j) = r \bmod 4$$

rotor

$$e(r, j) = (r + j) \bmod 4$$

Primary structural evaluation

Byte-local support proposition $\rightarrow$ DDT mean-probability null $\rightarrow$ model-exact
128-state weight-one transfer system $\rightarrow$ 256-context finite and periodic rates
with matched empirical diagnostics retained only as finite-resolution corroboration

Separate cross-byte boundary setting (not a one-variable width ablation).

The byte-local 8×8 binary layer is replaced by a 4×4 Cauchy MDS layer over $\text{GF}(2^8)$. Spatial width, field, and matrix family therefore change together. Detailed pruned-search outputs are reported in the supplement. They show only that schedule-dependent retained states can occur in this broader setting; no stable schedule ranking is claimed.